Torsional friction in planar contact

Sébastien Kleff
sk8001@nyu.edu

November 4, 2026

Abstract

Tactile sensors are widely used to detect slip, but much less is known about what can actually be inferred about frictional contact limits from tactile measurements. This question matters because detecting that slip is occurring or imminent is not the same as knowing which tangential and torsional loads the contact can still support. That distinction is important for control design. In this report, we study this question for a single planar contact patch under quasi-static Coulomb friction. We first show that coarse normal force information, consisting of the total normal force and center of pressure, is generally insufficient to determine the admissible tangential wrench set: two contacts can have the same total normal force and CoP while supporting different combinations of tangential force and torsional moment. We then study what can still be inferred from partial measurements. In particular, we derive upper and lower bounds on the admissible wrench set from coarse measurements and from spatially distributed taxel-wise normal forces. The latter provide tighter bounds by reducing the ambiguity on how the normal force is distributed over the contact patch. Taking the torques about a fixed point makes every bound a generalized moment problem: the bounds from taxel forces have closed forms, those combining taxel forces and CoP are an SOCP and an LP, and the gap between them halves, to first order, with each dyadic refinement of the taxel grid. Along a given load, the guaranteed and the possible wrench sets give a certified and an upper load factor, each a convex program, which decide whether the load is surely carried, surely not carried, or undecided by the measurement.

1 Introduction

Modern tactile sensors provide rich local information about contact pressure, shape, and deformation. Yet tactile features used for control are often selected mainly from empirical performance, for example contact pose, resultant force, image features, or local pressure measurements. A more basic question is less often made explicit: given a particular tactile measurement, what can actually be inferred about the physical interaction at the contact? We study this question in the setting of a rigid planar contact patch under Coulomb friction. Rather than trying to characterize every possible tactile signal or manipulation task, we focus on one concrete control-relevant quantity: the set of tangential forces and torsional moments that the contact can support without violating the friction model. The key difficulty is that this admissible wrench set depends on how the normal load is distributed over the patch. Coarse measurements such as the total normal force and center of pressure constrain this load distribution, but do not determine it uniquely. Spatial tactile measurements provide additional information and therefore reduce this ambiguity. The objective of this report is to make this relationship explicit. We ask **how the admissible frictional wrench set can be bounded from different levels of normal force information**, from total force and CoP to spatially distributed taxel-wise forces, **without requiring the full normal pressure field**.

2 Preliminaries (measures)

We briefly introduce the measure-theoretic notation used in the following. Only the notions needed for the contact model are recalled here; standard reference is [1].

2.1 Force as a measure.

Let $P \subset \mathbb{R}^2$ denote the contact patch, equipped with its Borel σ-algebra $\mathcal{B}(P)$. A *nonnegative measure* $\nu_N : \mathcal{B}(P) \to [0, +\infty]$ assigns to every measurable subset $A \subseteq P$ a nonnegative scalar $\nu_N(A)$, and is countably additive, i.e. for any pairwise disjoint measurable sets $A_i \subseteq P$,

$$\nu_N\left(\bigcup_i A_i\right) = \sum_i \nu_N(A_i). \tag{1}$$

In the present context, ν_N represents the (spatially distributed) normal force, i.e. $\nu_N(A)$ is the normal force carried by the region A. Hence $\nu_N(A)$ has units of force (N), and the resultant normal force is $f_N = \nu_N(P)$. We restrict attention to *finite nonnegative Radon measures*, denoted $\nu_N \in \mathcal{M}_+(P)$. Informally, a Radon measure is a "well-behaved" measure. In particular, on the compact sets considered here, it is finite and its value on measurable sets can be approximated from compact subsets. For our purpose, it essentially means that our measure does not assign an infinite force to P and it can represent either the force induced by an ordinary pressure field or an idealized concentrated point/line load.

We represent the tangential force by an $\mathbb{R}^2$-valued finite Radon measure $\nu_T = (\nu_{T,1}, \nu_{T,2})$ where each component $\nu_{T,i}$ is a finite signed Radon measure on P. Thus, for every measurable region $E \subseteq P$, the resultant tangential force carried by E is

$$\nu_T(E) = \begin{bmatrix} \nu_{T,1}(E) \\ \nu_{T,2}(E) \end{bmatrix} \in \mathbb{R}^2, \tag{2}$$

2.2 Area measure and pressure fields.

Let dA denote the usual area element on the planar patch P. A *normal pressure field* is a nonnegative function $\rho_N : P \to \mathbb{R}_+$ with units of force per unit area. If $\rho_N \in L^1_+(P)$, which simply means that ρ_N has finite integral over P

$$\rho_N(x) \geq 0 \quad \text{a.e.}, \qquad \int_P \rho_N(x)\, dA < \infty. \tag{3}$$

then it defines a finite normal force measure: the force carried by any region $E \subseteq P$ is given by

$$\nu_N(E) = \int_E \rho_N(x)\, dA. \tag{4}$$

which we also write equivalently as $d\nu_N(x) = \rho_N(x)\, dA$. In that case, ρ_N is called the *density* of the normal force measure ν_N with respect to area[1]. Not every normal force measure admits such a pressure field. For example the Dirac measure $\nu_N = f_N \delta_c$ represents an idealized normal force f_N concentrated at the single point c. It cannot be represented by an ordinary pressure field over area. In this note we adopt the measure-theoretic language (instead of only L^1_+ densities) so that we cover both "classical" pressure fields and idealized concentrated loads.

This notation allows ordinary pressure fields and concentrated loads to be handled by the same integration equations. For a measurable scalar or vector function g, the integral

$$\int_P g(x)\, d\nu_N(x) \tag{5}$$

[1] More generally, the Radon-Nikodym theorem guarantees the existence of a density for measures under suitable assumptions, such as "absolute continuity" which means physically that a subset of zero area (e.g. a point or a line) carries zero normal force. See [1], Theorem 1.30, p. 50

can be interpreted as a force-weighted spatial integral. Whenever ν_N admits a normal pressure field ρ_N,

$$\int_P g(x)\, d\nu_N(x) = \int_P g(x)\rho_N(x)\, dA. \tag{6}$$

For a Dirac measure,

$$\int_P g(x)\, d(f_N\delta_c)(x) = f_N g(c). \tag{7}$$

2.3 Moments of the normal force measure.

The zeroth moment of ν_N is its total measure,

$$m_0(\nu_N) \triangleq \int_P 1\, d\nu_N = \nu_N(P) = f_N. \tag{8}$$

In the present mechanical interpretation, this is the *resultant normal force*. The first moment is

$$m_1(\nu_N) \triangleq \int_P x\, d\nu_N(x) \in \mathbb{R}^2. \tag{9}$$

If $f_N > 0$, the *center of pressure* is the normalized first moment,

$$c[\nu_N] \triangleq \frac{m_1(\nu_N)}{m_0(\nu_N)} \tag{10}$$

$$= \frac{1}{f_N}\int_P x\, d\nu_N(x). \tag{11}$$

Hence knowing the total normal force and the center of pressure is equivalent to knowing the zeroth and first moments of the normal force measure $m_0(\nu_N) = f_N, m_1(\nu_N) = f_N c[\nu_N]$. Unlike the CoP, the pair (m_0, m_1) is linear in ν_N and defined for every measure, including $\nu_N = 0$. We work with these *homogeneous coordinates* throughout, and use the CoP only when $f_N > 0$.

Local moments. For a measurable subset $E \subseteq P$, such as a taxel, the *local first moment* of the force carried by E is $\int_E x\, d\nu_N(x)$. If $\nu_N(E) > 0$, the normalized first moment $\frac{1}{\nu_N(E)}\int_E x\, d\nu_N(x)$ is the *local CoP* of E, i.e., the barycenter of the restriction of ν_N to E; it lies in the closed convex hull of E.

3 Problem formulation

3.1 Planar contact model

Frame. All quantities are coordinates in a fixed frame $\mathcal{R} = (O, \mathcal{B})$ of the contact plane: O is the origin and $\mathcal{B} = (e_1, e_2)$ an orthonormal basis, completed by the contact normal e_3. A point of the plane is identified with its coordinate vector $x \in \mathbb{R}^2$. A wrench or a twist is always written in a frame, and its coordinates depend on the reference point (Lemma 3.4); here the reference point is O unless stated otherwise. For $x, y \in \mathbb{R}^2$, the *planar cross product* is the scalar

$$x \times y \triangleq \langle e_3, (x,0) \times (y,0)\rangle = x_1y_2 - x_2y_1 = (Jx)^\top y, \qquad J \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{12}$$

Let the planar contact patch $P \subset \mathbb{R}^2$ be compact, and let $\nu_N \in \mathcal{M}_+(P)$ and ν_T be the normal and tangential force measures introduced above. We do not assume $f_N > 0$. Convexity of P is assumed only where stated. We further assume a known and spatially uniform Coulomb friction

coefficient $\mu > 0$. We impose Coulomb friction in measure form by requiring that there exists a measurable density $\gamma_T : P \to \mathbb{R}^2$ of ν_T with respect to ν_N such that

$$d\nu_T(x) = \gamma_T(x)\, d\nu_N(x), \qquad \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e. on } P. \tag{13}$$

We call γ_T the *friction ratio field*. It can be interpreted as the tangential force per unit normal force. The above condition expresses both that tangential friction force can only be carried where normal force is present and that its local magnitude is bounded by μ[2]. When ν_N admits the normal pressure field ρ_N introduced above, i.e. $d\nu_N = \rho_N\, dA$, the tangential force measure also admits a density with respect to area,

$$\rho_T(x) = \gamma_T(x)\rho_N(x), \tag{14}$$

called the *tangential traction field*. The measure-form Coulomb condition then reduces to the classical local condition

$$\|\rho_T(x)\|_2 \le \mu\rho_N(x) \qquad \text{a.e. on } P. \tag{15}$$

We define the *resultant tangential force* and the *resultant torsional moment* about the origin O as

$$f_T \triangleq \nu_T(P) = \int_P \gamma_T(x)\, d\nu_N(x), \qquad \tau_N \triangleq \int_P x \times \gamma_T(x)\, d\nu_N(x). \tag{16}$$

Definition 3.1 (Admissible tangential wrench set)**.** *For a normal force measure $\nu_N \in \mathcal{M}_+(P)$, the admissible tangential wrench set is*

$$\mathcal{X}(\nu_N) \triangleq \left\{ (f_T, \tau_N) \in \mathbb{R}^2 \times \mathbb{R} : \begin{array}{l} \exists\, \gamma_T : P \to \mathbb{R}^2 \text{ measurable such that} \\ \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e.}, \\ f_T = \displaystyle\int_P \gamma_T(x)\, d\nu_N(x), \\ \tau_N = \displaystyle\int_P x \times \gamma_T(x)\, d\nu_N(x) \end{array} \right\}. \tag{17}$$

It is a nonempty, convex, compact subset of $\mathbb{R}^3$, symmetric about 0, containing all resultant tangential wrenches (tangential force and torsional moment about O) that are compatible with the normal force measure ν_N and the local Coulomb friction condition (13).

Proposition 3.2 (Support function of the admissible tangential wrench set)**.** *For any $\nu_N \in \mathcal{M}_+(P)$ and for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, the support function of $\mathcal{X}(\nu_N)$ is*

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P \|a + bJx\|_2 \; d\nu_N(x). \tag{18}$$

Proof. By definition, $\forall q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, $h_{\mathcal{X}(\nu_N)}(q) = \sup_{(f_T,\tau_N)\in\mathcal{X}(\nu_N)} \left(a^\top f_T + b\tau_N\right)$. For any friction ratio field γ_T,

$$a^\top f_T + b\tau_N = \int_P [a + bJx]^\top \gamma_T(x)\, d\nu_N(x) \le \mu \int_P \|a + bJx\|_2 \; d\nu_N(x), \tag{19}$$

by Cauchy–Schwarz and $\|\gamma_T(x)\|_2 \le \mu$. The bound is attained by the field of magnitude μ aligned with $a + bJx$ where this vector is nonzero, and $\gamma_T^\star(x) = 0$ elsewhere. □

[2]In particular, $d\nu_T = \gamma_T\, d\nu_N$ implies $\nu_T \ll \nu_N$. Conversely, by the Radon–Nikodym theorem, a tangential measure with $\|\nu_T(E)\|_2 \le \mu\, \nu_N(E)$ for every E has such a density.

Remark 3.3 (Twists and dissipated power)**.** *The direction $q = (a, b)$ is the coordinate vector of a planar rigid twist about O: a is the velocity of the material point at O and b the angular velocity, so the point at x has velocity $v_q(x) = a + bJx$. The power of the tangential forces under this twist is $\int_P \langle v_q(x), \gamma_T(x)\rangle \, d\nu_N(x) = a^\top f_T + b\tau_N = \langle q, (f_T, \tau_N)\rangle$. Hence $h_{\mathcal{X}(\nu_N)}(q)$ is the largest power that admissible friction can exert under the twist q; since $\mathcal{X}(\nu_N) = -\mathcal{X}(\nu_N)$, it is also the maximal power that friction can dissipate when the contact slides with twist q, which is the value selected by the maximal dissipation principle. This reading requires q and the wrench to be written about the same point.*

Lemma 3.4 (Change of reference point)**.** *For $o \in \mathbb{R}^2$, let $T_o(f, \tau) \triangleq (f, \tau - o \times f)$ and $S_o(a, b) \triangleq (a + bJo, b)$ (wrench and twist coordinates about o; Varignon's formula). Then T_o and S_o are invertible linear maps and*

(i) $\langle S_o q, T_o w\rangle = \langle q, w\rangle$ for all $q, w \in \mathbb{R}^3$: the power does not depend on the reference point;

(ii) the set of wrenches about o, $\mathcal{X}_o(\nu_N) \triangleq \{(f_T, \int_P (x - o) \times \gamma_T \, d\nu_N)\}$ (over the same fields γ_T), is $T_o\mathcal{X}(\nu_N)$, and

$$h_{\mathcal{X}_o(\nu_N)}(S_o q) = h_{\mathcal{X}(\nu_N)}(q), \qquad \text{i.e.} \qquad h_{\mathcal{X}_o(\nu_N)}(a', b) = \mu \int_P \|a' + bJ(x - o)\|_2 \, d\nu_N(x). \tag{20}$$

Proof. (i) $\langle S_o q, T_o w\rangle = (a + bJo)^\top f + b(\tau - o \times f) = a^\top f + b\tau$, since $o \times f = (Jo)^\top f$. (ii) $\int_P (x - o) \times \gamma_T \, d\nu_N = \tau_N - o \times f_T$. Then $h_{\mathcal{X}_o(\nu_N)}(S_o q) = \sup_{w \in \mathcal{X}(\nu_N)} \langle S_o q, T_o w\rangle = h_{\mathcal{X}(\nu_N)}(q)$ by (i), and $S_o^{-1}(a', b) = (a' - bJo, b)$ gives the integral form. □

Remark 3.5 (Torsion about the CoP)**.** *For $f_N > 0$, the set $\mathcal{W}(\nu_N) \triangleq \mathcal{X}_{c[\nu_N]}(\nu_N)$ of wrenches about the CoP is the one usually considered for torsional friction (soft-finger models), with support function $\mu \int_P \|a + bJ(x - c[\nu_N])\|_2 \, d\nu_N$. For a single measure the two descriptions are equivalent. When several measures are combined, as in the bounds below, the change of coordinates $T_{c[\nu_N]}$ depends on the measure: unions, intersections, infima and suprema of $\mathcal{W}(\nu_N)$ over measures with different CoPs combine coordinates written about different points, which no longer describe one physical twist or one set of physical wrenches. We therefore work with $\mathcal{X}$ throughout and recover statements about the CoP through Lemma 3.4 when the CoP is measured.*

Lemma 3.6 (Linearity in the normal force)**.** *For $\nu_1, \nu_2 \in \mathcal{M}_+(P)$ and $t \geq 0$, $h_{\mathcal{X}(\nu_1 + \nu_2)} = h_{\mathcal{X}(\nu_1)} + h_{\mathcal{X}(\nu_2)}$ and $h_{\mathcal{X}(t\nu_1)} = t\, h_{\mathcal{X}(\nu_1)}$. Consequently $\mathcal{X}(\nu_1 + \nu_2) = \mathcal{X}(\nu_1) \oplus \mathcal{X}(\nu_2)$ (Minkowski sum), $\mathcal{X}(t\nu_1) = t\, \mathcal{X}(\nu_1)$ and $\mathcal{X}(0) = \{0\}$.*

Proof. The first claims follow from Proposition 3.2. The support function of a Minkowski sum is the sum of the support functions, and nonempty compact convex sets with the same support function are equal [4, Section 13]. □

3.2 Measurement-conditioned support bounds

We focus here on the case where we do not know the *true* force measure $\nu_N^\star$ but we have only access to some measurements of it (e.g. resultant normal force, CoP, etc.). The goal is to investigate what can be known about the support of $\mathcal{X}(\nu_N^\star)$ given partial measurements of $\nu_N^\star$.

Symbol	Meaning	Remark	Lives in
ν_N	normal force measure	true $\nu_N^\star$ is unknown	$\mathcal{M}_+(P)$
(m_0, m_1)	zeroth and first moments of ν_N	$m_1 = f_N c[\nu_N]$ if $f_N > 0$	$\mathbb{R} \times \mathbb{R}^2$
$q = (a, b)$	twist about O (support direction)	prescribed	$\mathbb{R}^2 \times \mathbb{R}$
$\mathcal{M}$	measurement operator	fixed by the sensing model	$\mathcal{M}_+(P) \to \mathbb{R}^r$
y	measurement value	observed data from $(\mathcal{M}, \nu_N)$	$\mathbb{R}^r$
$\Pi_{\mathcal{M}}(y)$	all measures compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq \mathcal{M}_+(P)$
$\underline{h}(q; y, \mathcal{M})$	measurement-conditioned lower bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\overline{h}(q; y, \mathcal{M})$	measurement-conditioned upper bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\mathcal{X}^-(y, \mathcal{M})$, $\mathcal{X}^+(y, \mathcal{M})$	guaranteed and possible wrench sets	Proposition 3.11	$\subseteq \mathbb{R}^3$

Table 1: Notations.

Definition 3.7 (Measurement operator)**.** *A measurement operator is a map $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ of the form*

$$\mathcal{M}(\nu_N) = \begin{bmatrix} \int_P \phi_1(x)\, d\nu_N(x) \\ \vdots \\ \int_P \phi_r(x)\, d\nu_N(x) \end{bmatrix}, \qquad \forall \nu_N \in \mathcal{M}_+(P), \tag{21}$$

where the prescribed bounded Borel functions $\phi_j : P \to \mathbb{R}$ are called the measurement functions. In the following, we denote a measurement value (or observation) induced by the (unknown) true force measure $\nu_N^\star$ as $y = \mathcal{M}(\nu_N^\star) \in \mathbb{R}^r$.

Definition 3.8 (Measurement-compatible normal force family)**.** *Given a measurement operator $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ and the observation $y \in \mathbb{R}^r$, the set of normal force measures compatible with y is the preimage*

$$\Pi_{\mathcal{M}}(y) \triangleq \{\nu_N \in \mathcal{M}_+(P) : \mathcal{M}(\nu_N) = y\} = \mathcal{M}^{-1}[\{y\}]. \tag{22}$$

It contains all the normal force measures that are indistinguishable under the measurement y, and it is convex.

Assumption 3.9 (Mass control)**.** *There exists $\lambda \in \mathbb{R}^r$ such that $\sum_j \lambda_j \phi_j(x) \geq 1$ for every $x \in P$.*

Under Assumption 3.9, every $\nu_N \in \Pi_{\mathcal{M}}(y)$ satisfies $\nu_N(P) \leq \int_P \sum_j \lambda_j \phi_j \, d\nu_N = \lambda^\top y$, so the compatible family has bounded mass. All measurement operators studied below measure the resultant normal force, directly or as the sum of the taxel forces, and satisfy it. Without it, force could be hidden where all the measurement functions vanish, and the upper bound below would be infinite. No positivity assumption on f_N is needed.

Given an observation y of the unknown true force measure $\nu_N^\star$, we can compute measurement-conditioned lower/upper bounds on the support function $h_{\mathcal{X}(\nu_N^\star)}$ by optimizing over $\Pi_{\mathcal{M}}(y)$.

Definition 3.10 (Measurement-conditioned support bounds)**.** *Let $\mathcal{M}$ be a measurement operator and $y \in \mathbb{R}^r$ an observation. For a twist $q \in \mathbb{R}^2 \times \mathbb{R}$, the measurement-conditioned bounds in direction q are*

$$\underline{h}(q; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{23}$$

$$\overline{h}(q; y, \mathcal{M}) \triangleq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{24}$$

and the guaranteed and possible wrench sets are

$$\mathcal{X}^-(y, \mathcal{M}) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N), \qquad \mathcal{X}^+(y, \mathcal{M}) \triangleq \bigcup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N). \tag{25}$$

The objective of both problems in (23) is linear in ν_N, and the constraints $\int_P \phi_j \, d\nu_N = y_j$ are linear moment constraints: *the bounds are Generalized Moment Problems (GMP)* [2], for every measurement operator. This is a consequence of the fixed reference point; about the CoP of each measure the objective would depend on ν_N through $c[\nu_N]$ (Remark 3.5).

Proposition 3.11 (Guaranteed and possible wrench sets)**.** *Let $\mathcal{M}$ satisfy Assumption 3.9 and $y \in \mathbb{R}^r$.*

(i) The possible set $\mathcal{X}^+(y, \mathcal{M})$ is convex, and its support function is $\overline{h}(\cdot; y, \mathcal{M})$.

(ii) The guaranteed set is $\mathcal{X}^-(y, \mathcal{M}) = \{w : \langle q, w\rangle \le \underline{h}(q; y, \mathcal{M}) \ \forall q\}$. Its support function is the closed convex envelope of $\underline{h}(\cdot; y, \mathcal{M})$, which is in general strictly below $\underline{h}$.

(iii) $y \mapsto \overline{h}(q; y, \mathcal{M})$ is concave and $y \mapsto \underline{h}(q; y, \mathcal{M})$ is convex on the set of observations $\mathcal{M}(\mathcal{M}_+(P))$, and both are positively homogeneous.

Proof. (i) Let $w_k \in \mathcal{X}(\nu_k)$ with $\nu_k \in \Pi_{\mathcal{M}}(y)$, $k = 1, 2$, and $t \in [0, 1]$. By Lemma 3.6, $tw_1 + (1-t)w_2 \in t\mathcal{X}(\nu_1) \oplus (1-t)\mathcal{X}(\nu_2) = \mathcal{X}(t\nu_1 + (1-t)\nu_2)$, and $t\nu_1 + (1-t)\nu_2 \in \Pi_{\mathcal{M}}(y)$ since $\Pi_{\mathcal{M}}(y)$ is convex. The support function of a union is the supremum of the support functions. (ii) $w \in \mathcal{X}^-$ if and only if $w \in \mathcal{X}(\nu_N)$ for every compatible ν_N, i.e., $\langle q, w\rangle \le h_{\mathcal{X}(\nu_N)}(q)$ for all q and all compatible ν_N [4, Section 13]. The support function of the set $\{w : \langle q, w\rangle \le \underline{h}(q) \ \forall q\}$ is the closed convex envelope of $\underline{h}$ [4, Section 13]. (iii) If $\nu_k \in \Pi_{\mathcal{M}}(y_k)$, then $t\nu_1 + (1-t)\nu_2 \in \Pi_{\mathcal{M}}(ty_1 + (1-t)y_2)$, and $\nu_N \mapsto h_{\mathcal{X}(\nu_N)}(q)$ is linear; take the supremum (resp. infimum) over ν_1, ν_2. □

Property (i) depends on the fixed reference point: with the CoP-referenced sets $\mathcal{W}(\nu_N)$, a convex combination of measures with different CoPs moves the reference point, and the union need not be convex. Property (iii) generalizes the concavity of Proposition 4.4 of Yann's draft (Remark 4.6).

Remark 3.12 (How to use the bounds)**.** *A candidate tangential wrench w (about O) is carried by every compatible normal force measure, i.e., $w \in \mathcal{X}^-(y, \mathcal{M})$, if and only if $\langle q, w\rangle \le \underline{h}(q; y, \mathcal{M})$ for all q. Conversely, if $\langle q, w\rangle > \overline{h}(q; y, \mathcal{M})$ for some q, then $w \notin \mathcal{X}^+(y, \mathcal{M})$: no compatible measure carries w, and the contact slips whatever the actual pressure distribution. By Proposition 3.11(i), the second test is a convex membership test. Section 9 turns both tests into convex programs along a load direction*[3]*.*

3.3 General structure of the bounds

For $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, we write

$$g(x; q) \triangleq \|a + bJx\|_2 \,, \tag{26}$$

the speed of the material point at x under the twist q, so that, by Proposition 3.2, $h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P g(x; q) \, d\nu_N(x)$. For $b \neq 0$, $a + bJx = bJ(x - p)$ with $p \triangleq Ja/b$, the instantaneous center of rotation, so that $g(x; q) = |b| \, \|x - p\|_2$. For $b = 0$, $g(x; q) = \|a\|_2$.

Lemma 3.13 (Properties of the integrand)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$:*

(i) $x \mapsto g(x; q)$ is convex on $\mathbb{R}^2$;

(ii) for all $x, x' \in \mathbb{R}^2$, $|g(x; q) - g(x'; q)| \le |b| \, \|x - x'\|_2$;

[3] Because all support functions are positively homogeneous in q, it is sufficient to consider one normalized representative of each nonzero ray.

(iii) for every $\nu_N \in \mathcal{M}_+(P)$,

$$h_{\mathcal{X}(\nu_N)}(q) \geq \mu \left\| m_0(\nu_N)\, a + bJm_1(\nu_N) \right\|_2, \tag{27}$$

which, for $f_N > 0$, reads $h_{\mathcal{X}(\nu_N)}(q) \geq \mu f_N\, g(c[\nu_N]; q)$: the maximal dissipated power is at least that of the whole normal force concentrated at the CoP.

Proof. (i) $g(\cdot; q)$ is a norm composed with an affine map. (ii) By the reverse triangle inequality and $\|J\|_2 = 1$. (iii) By the triangle inequality for vector-valued integrals, $\int_P \|a + bJx\|_2 \, d\nu_N \geq \| \int_P (a + bJx)\, d\nu_N \|_2 = \|m_0 a + bJm_1\|_2$. □

Comparing measurement operators. The bounds of two measurement operators can only be compared if their observations come from the same normal force measure. This leads to an order on the information carried by measurement operators.

Definition 3.14 (Information preorder)**.** *A measurement operator $\mathcal{M}$ is at least as informative as $\mathcal{M}'$, written $\mathcal{M} \succeq \mathcal{M}'$, if*

$$\Pi_{\mathcal{M}}(\mathcal{M}(\nu_N)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\nu_N)) \qquad \text{for every } \nu_N \in \mathcal{M}_+(P), \tag{28}$$

i.e., every measure that $\mathcal{M}$ cannot distinguish from ν_N is also indistinguishable from ν_N under $\mathcal{M}'$. The concatenation $\mathcal{M} \vee \mathcal{M}' \triangleq (\mathcal{M}, \mathcal{M}')$ is the operator with measurement functions $(\phi_1, \ldots, \phi_r, \phi'_1, \ldots, \phi'_{r'})$.

Proposition 3.15 (Properties of the information preorder)**.** *(i) $\succeq$ is a preorder (reflexive and transitive). It is not antisymmetric, and not total.*

(ii) $\mathcal{M} \succeq \mathcal{M}'$ if and only if there is a matrix $A \in \mathbb{R}^{r' \times r}$ with $\phi'_j(x) = \sum_k A_{jk} \phi_k(x)$ for every $x \in P$ and every j, i.e., $\mathcal{M}' = A\,\mathcal{M}$.

(iii) $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}$ and $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}'$, with $\Pi_{\mathcal{M} \vee \mathcal{M}'}((y, y')) = \Pi_{\mathcal{M}}(y) \cap \Pi_{\mathcal{M}'}(y')$.

Proof. (i) Inclusion is reflexive and transitive. Scaling the measurement functions by 2 gives a different operator with the same compatible families, and the operators $\mathcal{M}_0$ and $\mathcal{M}_F$ of the next sections are not comparable (see (ii)). (ii) If $\mathcal{M}' = A\mathcal{M}$, then $\mathcal{M}(\nu) = \mathcal{M}(\nu_N)$ implies $\mathcal{M}'(\nu) = \mathcal{M}'(\nu_N)$. Conversely, write $\Phi(x) = (\phi_k(x))_k$ and $\Phi'(x) = (\phi'_j(x))_j$. A linear map A with $A\Phi(x) = \Phi'(x)$ for all $x \in P$ exists if and only if $\sum_i \alpha_i \Phi(x_i) = 0$ implies $\sum_i \alpha_i \Phi'(x_i) = 0$ for all finite families $x_i \in P$, $\alpha_i \in \mathbb{R}$. Given such a family, let $\sigma^{\pm} = \sum_{i:\pm\alpha_i > 0} |\alpha_i| \delta_{x_i} \in \mathcal{M}_+(P)$. Then $\mathcal{M}(\sigma^+) = \mathcal{M}(\sigma^-)$, so $\sigma^- \in \Pi_{\mathcal{M}}(\mathcal{M}(\sigma^+)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\sigma^+))$, i.e., $\sum_i \alpha_i \Phi'(x_i) = \mathcal{M}'(\sigma^+) - \mathcal{M}'(\sigma^-) = 0$. (iii) Immediate from the definitions. □

The characterization (ii) is pointwise: functions that agree almost everywhere with respect to area are not equivalent, since point loads see their values at single points. The number r of measurement functions is not monotone with respect to $\succeq$, because measurement functions can be redundant.

Theorem 3.16 (Monotonicity with respect to information)**.** *Let $\mathcal{M} \succeq \mathcal{M}'$, $\nu_N \in \mathcal{M}_+(P)$, $y = \mathcal{M}(\nu_N)$ and $y' = \mathcal{M}'(\nu_N)$. Then, for every $q \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\underline{h}(q; y', \mathcal{M}') \leq \underline{h}(q; y, \mathcal{M}) \leq h_{\mathcal{X}(\nu_N)}(q) \leq \overline{h}(q; y, \mathcal{M}) \leq \overline{h}(q; y', \mathcal{M}'), \tag{29}$$

and $\mathcal{X}^-(y', \mathcal{M}') \subseteq \mathcal{X}^-(y, \mathcal{M}) \subseteq \mathcal{X}(\nu_N) \subseteq \mathcal{X}^+(y, \mathcal{M}) \subseteq \mathcal{X}^+(y', \mathcal{M}')$. This applies in particular to the true measure $\nu_N^\star$.

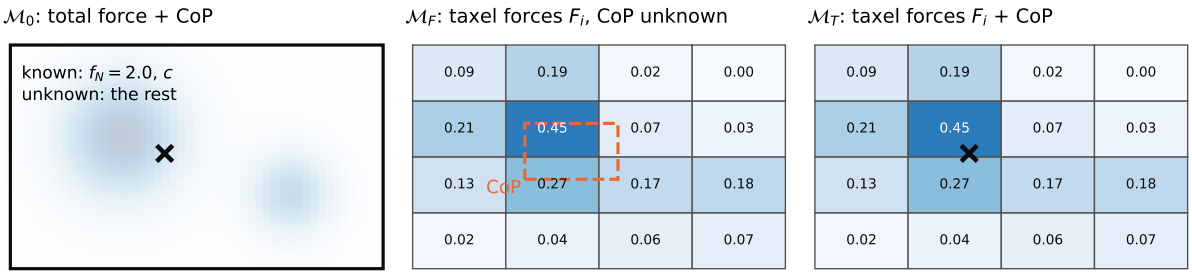


Figure 1: What each measurement operator observes for the same contact (4×4 taxels). $\mathcal{M}_0$: only f_N and the CoP, the pressure distribution (faded) is unknown. $\mathcal{M}_F$: the taxel forces F_i (cells filled with the mean pressure $F_i/|P_i|$), the CoP is only known to lie in the weighted Minkowski average of the taxels (dashed, Remark 5.9). $\mathcal{M}_T$: taxel forces and CoP.

Measurement	Observed data	Lower bound	Upper bound
$\mathcal{M}_0$ (Sec. 4)	$y_0 = (f_N, m_1)$	closed form	LP over the vertices of P (closed form via triangles)
$\mathcal{M}_F$ (Sec. 6)	$y_F = (F_1, \dots, F_r)$	closed form: $\sum_i$ of minima over the taxels	closed form: $\sum_i$ of maxima over the taxel corners
$\mathcal{M}_T$ (Sec. 7)	$y_T = (y_0, y_F)$	SOCP	LP over the taxel corners

Table 2: Measurement operators and structure of the resulting bounds. All bounds are GMPs; the CoP constraint of $\mathcal{M}_T$ couples the taxels.

Proof. $\nu_N \in \Pi_{\mathcal{M}}(y) \subseteq \Pi_{\mathcal{M}'}(y')$. An infimum (resp. supremum, intersection, union) over a smaller family is larger (resp. smaller, larger, smaller). □

Remark 3.17 (Attainment)**.** *The bounds are defined as an infimum and a supremum, and we do not need them to be attained: only their values enter the tests of Remark 3.12 and the monotonicity of Theorem 3.16. For the taxel measurements, the finite-dimensional formulas below are attained by measures that may put force on a taxel boundary owned by a neighboring taxel; Lemma 5.6 shows that they are nevertheless the exact values of the bounds.*

Table 2 summarizes the measurement operators studied in the remainder of the report and the structure of the resulting problems, and Figure 1 illustrates what each of them observes. We next specialize (23) by increasing information: the coarse measurement (Section 4), the taxel forces (Section 6), and both (Section 7). The tightening of the bounds as information is added is studied in Section 8.

4 $\mathcal{M}_0$ case (coarse measurement)

We first assume that the available measurements consist only of the resultant normal force and its first moment, i.e., the CoP when $f_N > 0$. **Known:** P, μ, f_N and m_1. **Unknown:** how $\nu_N^\star$ is distributed over P. In this section P is convex.

Definition 4.1 (Coarse normal force measurement)**.** *Given the true (unknown) normal force measure $\nu_N^\star \in \mathcal{M}_+(P)$, the coarse measurement is its zeroth and first moments*

$$y_0 = \mathcal{M}_0(\nu_N^\star) \triangleq \left(\int_P d\nu_N^\star(x),\ \int_P x\, d\nu_N^\star(x) \right) = (f_N, m_1). \tag{30}$$

Its measurement functions are $\phi_1(x) = 1$, $\phi_2(x) = x_1$, $\phi_3(x) = x_2$, and it satisfies Assumption 3.9 with $\lambda = (1, 0, 0)$. For $f_N > 0$, y_0 is equivalent to (f_N, c) with $c = m_1/f_N \in P$.

Theorem 4.2 (Exact coarse lower bound)**.** *Given the coarse observation $y_0 = (f_N, m_1)$, for any $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_0, \mathcal{M}_0) = \mu \, \|f_N a + bJm_1\|_2} \tag{31}$$

For $f_N > 0$ this is $\mu f_N \, g(c; q)$, the maximal dissipated power of the normal force concentrated at the CoP.

Proof. By Lemma 3.13(iii), $h_{\mathcal{X}(\nu_N)}(q) \geq \mu \|f_N a + bJm_1\|_2$ for every $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. If $f_N = 0$, then $\Pi_{\mathcal{M}_0}(y_0) = \{0\}$ and both sides vanish. Otherwise $c \in P$, so $f_N \delta_c \in \Pi_{\mathcal{M}_0}(y_0)$, and it attains the value. □

Corollary 4.3 (Exact guaranteed set; no torsion about the CoP)**.** *Let $f_N > 0$ and $c = m_1/f_N$. Then*

$$\mathcal{X}^-(y_0, \mathcal{M}_0) = \mathcal{X}(f_N \delta_c) = \{(f, c \times f) : \ \|f\|_2 \leq \mu f_N\}. \tag{32}$$

In coordinates about the CoP (Lemma 3.4), $T_c \mathcal{X}^-(y_0, \mathcal{M}_0) = f_N \, B(0, \mu) \times \{0\}$: the coarse measurement guarantees the friction disc for the tangential force, but no torsional moment about the CoP. Equivalently, for the twists rotating about the CoP, $q = (-bJc, b)$, $\underline{h}(q; y_0, \mathcal{M}_0) = 0$.

Proof. For the point load, a field is a single vector $\gamma \in B(0, \mu)$, with resultant $(f_N \gamma, \, c \times f_N \gamma)$, which gives the second equality, and $\mathcal{X}^- \subseteq \mathcal{X}(f_N \delta_c)$ since $f_N \delta_c$ is compatible. Conversely, let $\|f\|_2 \leq \mu f_N$ and $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. The uniform field $\gamma_T \equiv f/f_N$ is admissible, with $f_T = f$ and $\tau_N = \int_P x \times (f/f_N) \, d\nu_N = m_1 \times f/f_N = c \times f$ (compare Remark 4.2 of Yann's draft). □

Theorem 4.4 (Coarse upper bound)**.** *The supremum $\overline{h}(q; y_0, \mathcal{M}_0)$ is attained by a measure $\nu_N^+ \in \Pi_{\mathcal{M}_0}(y_0)$ supported on at most three points, i.e.,*

$$\nu_N^+ = \sum_{j=1}^{s} \lambda_j \delta_{x_j}, \qquad s \leq 3, \qquad x_j \in P, \quad \lambda_j \geq 0, \qquad \sum_{j=1}^{s} \lambda_j = f_N, \qquad \sum_{j=1}^{s} \lambda_j x_j = m_1. \tag{33}$$

Proof. This is Proposition 4.4 of Yann's draft with the Coulomb set $C = B(0, \mu)$ (Remark 4.6), where it follows from the fact that the extreme points of a set of measures defined by three linear moment constraints are supported on at most three points [3, 2]. A finite-dimensional argument is as follows. If $f_N = 0$ there is nothing to prove. Otherwise, the constraints and the objective depend on $\pi = \nu_N / f_N$ only through its moment vector $\int_P \Phi \, d\pi$, with $\Phi(x) = (x, g(x; q)) \in \mathbb{R}^3$, which lies in the compact convex set $K = \operatorname{conv} \Phi(P)$. The bound is $\mu f_N t^\star$, where $(c, t^\star)$ is the highest point of K above c. This point lies on a supporting plane H of K, and $K \cap H = \operatorname{conv}(\Phi(P) \cap H)$, so Carathéodory's theorem in the plane H writes it as a convex combination $\sum_j \beta_j \Phi(x_j)$ of $s \leq 3$ points. The measure $f_N \sum_j \beta_j \delta_{x_j}$ is compatible and attains the bound. □

Corollary 4.5 (Polygonal contact patch)**.** *Suppose additionally that P is a convex polygon with vertices $v_1, \ldots, v_K$.*

(i) The coarse upper bound is the linear program

$$\boxed{\begin{aligned} \overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\lambda_1, \ldots, \lambda_K \geq 0} \quad & \sum_{k=1}^{K} \lambda_k \|a + bJv_k\|_2 \\ s.t. \quad & \sum_{k=1}^{K} \lambda_k = f_N, \qquad \sum_{k=1}^{K} \lambda_k v_k = m_1, \end{aligned}} \tag{34}$$

which has an optimal solution with at most three nonzero weights.

(ii) For a vertex triangle $\Delta = \operatorname{conv}\{v_i, v_j, v_k\}$ *containing* c*, let* $\lambda^{\Delta}(y_0) \in \mathbb{R}^3_+$ *be the unique weights with* $\sum \lambda^{\Delta} = f_N$*,* $\sum \lambda^{\Delta}_l v_l = m_1$ *(homogeneous barycentric coordinates, linear in* y_0*). Then*

$$\overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\Delta \ni c} \sum_{l \in \Delta} \lambda^{\Delta}_l(y_0)\, \|a + bJv_l\|_2, \tag{35}$$

the maximum over the vertex triangles containing c*. If* P *is a triangle,* $\overline{h}(q; \cdot, \mathcal{M}_0)$ *is linear in* y_0*. If* P *is a convex quadrilateral (e.g. a rectangle), each of its two diagonal triangulations has a triangle containing* c*, and the bound is the larger of the corresponding values.*

Proof. (i) By Theorem 4.4, an optimal measure is $\sum_{j=1}^{s} \lambda_j \delta_{x_j}$ with $s \le 3$. Write each x_j as a convex combination $\sum_k \alpha_{jk} v_k$ of the vertices. Moving the mass $\lambda_j \alpha_{jk}$ from x_j to v_k preserves the zeroth and first moments and, by convexity of $g(\cdot; q)$ (Lemma 3.13(i)), does not decrease the objective. This is the reduction to the extreme points of P in Yann's Proposition 4.4. Hence an optimal measure supported on the vertices exists, which gives the LP. The LP has three equality constraints, so it has a basic optimal solution with at most three nonzero weights. (ii) The basic feasible solutions are supported on affinely independent vertices whose convex hull contains c; completing them to a vertex triangle containing c (with zero weights) gives the weights $\lambda^{\Delta}(y_0)$, which are unique since the vertices of Δ are affinely independent. An LP attains its maximum at a basic solution. The diagonal triangulations of a quadrilateral cover it. □

Remark 4.6 (Connection with Yann's formulation)**.** *Theorem 4.4 and Corollary 4.5 are the Coulomb case* $C = B(0, \mu)$ *of Proposition 4.4 in Yann's draft, with the same reference point. With his notation* $x^{\perp} = -Jx$*, his moment* $m_N = -\int \langle \rho_T, x^{\perp} \rangle$ *is our* τ_N*, his twist* (ω_N, v_T) *is our* $q = (v_T, \omega_N)$*, and his integrand is* $s_C(\omega_N x^{\perp} - v_T) = \mu \|v_T + \omega_N Jx\|_2 = \mu\, g(x; q)$*. His set* $f_N\, D(c^{\perp})$ *is* $\mathcal{X}^+(y_0, \mathcal{M}_0)$ *(components reordered), so that* $f_N\, s_{D(c^{\perp})}(-[\omega_N, v_T]) = \overline{h}(q; y_0, \mathcal{M}_0)$*, and the concavity of* $a \mapsto s_{D(a)}$ *is Proposition 3.11(iii). His special cases are the CoP at a vertex of* P *(a single point load, both bounds equal* $\mu f_N g(c; q)$*) and* $\omega_N = 0$ *(both bounds equal* $\mu f_N \|v_T\|_2$*). As he notes,* $D(\cdot)$ *is the place to add priors on the distribution; the taxel measurements of the next sections are such priors.*

The two bounds differ in general, which shows that the coarse measurement does not determine the admissible wrench set.

Example 4.7 (Segment patch)**.** *Let* $P = [0, L] \times \{0\}$*,* $c = \left(\frac{L}{2}, 0\right)$*,* $\nu_N^{(1)} = f_N \delta_c$ *and* $\nu_N^{(2)} = \frac{f_N}{2}\delta_{(0,0)} + \frac{f_N}{2}\delta_{(L,0)}$*. Both have* $\mathcal{M}_0(\nu_N^{(1)}) = \mathcal{M}_0(\nu_N^{(2)}) = \left(f_N, f_N \left(\frac{L}{2}, 0\right)\right)$*. For the unit rotation about the CoP,* $q_c = (-Jc, 1)$*, so that* $g(x; q_c) = \|x - c\|_2$*,*

$$h_{\mathcal{X}(\nu_N^{(1)})}(q_c) = 0 \quad \neq \quad h_{\mathcal{X}(\nu_N^{(2)})}(q_c) = \mu f_N \frac{L}{2} = \overline{h}(q_c; y_0, \mathcal{M}_0). \tag{36}$$

The results of this section are recovered in Section 7 as the special case of a single taxel $P_1 = P$.

5 Taxel measurements: preliminaries (common setting)

We now assume that a tactile array measures the normal force carried by each taxel, with or without the coarse measurement. This section fixes the setting and collects the lemmas used in Sections 6 and 7.

Assumption 5.1 (Taxel partition)**.** *The patch* P *is partitioned into* r *nonempty, pairwise disjoint, convex Borel sets* $P_1, \dots, P_r$*, called taxels, with* $P = \bigcup_{i=1}^{r} P_i$*. When stated, the closure of each taxel is a convex polygon* $\overline{P_i} = \operatorname{conv}\{v_{i1}, \dots, v_{iK_i}\}$ *with vertices* $v_{ik} \in \mathbb{R}^2$*.*

Definition 5.2 (Dyadic taxel grids)**.** *Let $P = [0, L] \times [0, \ell]$. For $p_x, p_y \in \mathbb{N}$, the grid $G(p_x, p_y)$ has the $2^{p_x} \times 2^{p_y}$ half-open cells $[k\frac{L}{2^{p_x}}, (k+1)\frac{L}{2^{p_x}}) \times [l\frac{\ell}{2^{p_y}}, (l+1)\frac{\ell}{2^{p_y}})$, the cells of the last column and row being closed on their outer side, so that they partition P. Each cell is a rectangle of size $w_x \times w_y = \frac{L}{2^{p_x}} \times \frac{\ell}{2^{p_y}}$, and $G(p_x, p_y)$ satisfies Assumption 5.1.*

Definition 5.3 (Taxel force measurement)**.** *Given the true unknown normal force measure $\nu_N^\star$ and the partition $\{P_i\}_{i=1}^r$, the taxel force measurement is*

$$y_F = \mathcal{M}_F(\nu_N^\star) \triangleq (\nu_N^\star(P_1), \dots, \nu_N^\star(P_r)), \tag{37}$$

with measurement functions $\phi_i = \mathbf{1}_{P_i}$, $i = 1, \dots, r$. The measured taxel forces are denoted $F_i \triangleq \nu_N^\star(P_i)$, and $f_N = \sum_i F_i$. Since $\sum_i \mathbf{1}_{P_i} = 1$, $\mathcal{M}_F$ satisfies Assumption 3.9. For the dyadic grid $G(p_x, p_y)$ we write $\mathcal{M}_F(p_x, p_y)$ for this operator (Yann's family $\mathcal{M}(T; p_x, p_y)$).

Definition 5.4 (Combined coarse and taxel force measurement)**.** *The combined measurement is the concatenation $\mathcal{M}_T \triangleq \mathcal{M}_0 \vee \mathcal{M}_F$,*

$$y_T = \mathcal{M}_T(\nu_N^\star) = (f_N, m_1, F_1, \dots, F_r), \qquad \Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F). \tag{38}$$

For the dyadic grid $G(p_x, p_y)$ we write $\mathcal{M}_T(p_x, p_y) \triangleq \mathcal{M}_0 \vee \mathcal{M}_F(p_x, p_y)$ (Yann's family $\mathcal{M}(T_c; p_x, p_y)$).

Proposition 5.5 (Information chains)**.** *(i) If the partition $\{P'_j\}$ refines $\{P_i\}$ (each P_i is a union of cells P'_j), then $\mathcal{M}'_F \succeq \mathcal{M}_F$ and $\mathcal{M}'_T \succeq \mathcal{M}_T$. In particular, $\mathcal{M}_F(p'_x, p'_y) \succeq \mathcal{M}_F(p_x, p_y)$ and $\mathcal{M}_T(p'_x, p'_y) \succeq \mathcal{M}_T(p_x, p_y)$ whenever $p'_x \geq p_x$ and $p'_y \geq p_y$, with $\mathcal{M}_F(0,0)$ the resultant normal force alone and $\mathcal{M}_T(0,0)$ equivalent to $\mathcal{M}_0$ (each $\succeq$ the other).*

(ii) $\mathcal{M}_T \succeq \mathcal{M}_0$ and $\mathcal{M}_T \succeq \mathcal{M}_F$. If $r \geq 2$ and the taxels have nonempty interiors, $\mathcal{M}_0$ and $\mathcal{M}_F$ are not comparable.

(iii) Grids that are not nested are not comparable in general: e.g. 3×3 and 4×4 grids of the same rectangle, or $G(p'_x, p'_y)$ and $G(p_x, p_y)$ with $p'_x > p_x$ and $p'_y < p_y$.

Proof. By Proposition 3.15(ii). (i) $\mathbf{1}_{P_i} = \sum_{j: P'_j \subseteq P_i} \mathbf{1}_{P'_j}$ pointwise, and concatenation with $\mathcal{M}_0$ preserves the order. (ii) Proposition 3.15(iii); an indicator of a proper taxel is not affine on P, and x_1 is not constant on a taxel with nonempty interior. (iii) An indicator of a cell of one grid is not constant on some cell of the other. □

Taxel boundaries. Because the taxels are half-open, a measure that puts force on the boundary of a taxel counts it in one taxel only. The finite-dimensional formulas below are attained by measures that may put force on a corner owned by a neighboring taxel. We therefore introduce the *closure model*

$$\overline{\Pi}_{\mathcal{M}_F}(y_F) \triangleq \Big\{ \nu_N = \sum_{i=1}^r \nu_i : \ \nu_i \in \mathcal{M}_+(P) \text{ supported on } \overline{P_i}, \ \nu_i(\overline{P_i}) = F_i \Big\}, \tag{39}$$

$$\overline{\Pi}_{\mathcal{M}_T}(y_T) \triangleq \overline{\Pi}_{\mathcal{M}_F}(y_F) \cap \Pi_{\mathcal{M}_0}(y_0), \tag{40}$$

which contains the compatible families (take ν_i the restriction of ν_N to P_i), and show that it gives the same bounds.

Lemma 5.6 (Half-open taxels)**.** *Let the taxels be the cells of a dyadic grid (Definition 5.2). For $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, an observation $y = \mathcal{M}(\nu_N^\star)$ and every q, the infimum and the supremum of $h_{\mathcal{X}(\nu_N)}(q)$ over $\Pi_{\mathcal{M}}(y)$ and over $\overline{\Pi}_{\mathcal{M}}(y)$ coincide.*

Proof. One inclusion is immediate. Conversely, let $\nu_N = \sum_i \nu_i \in \overline{\Pi}_{\mathcal{M}}(y)$ and $\varepsilon \in (0,1]$. For each i with $F_i > 0$, let $p_i \in P_i$ be the local CoP of $\nu_N^\star$ in P_i, which lies in the half-open rectangle P_i (an average of points of $[\alpha, \beta)$ lies in $[\alpha, \beta)$), and push every mass of ν_i a fraction ε toward it: ν_i^ε is the image of ν_i under $x \mapsto (1-\varepsilon)x + \varepsilon p_i$. For a rectangle this map sends $\overline{P_i}$ into P_i coordinate-wise, so $\nu_N^\varepsilon = \sum_i \nu_i^\varepsilon$ has taxel forces F_i. Its first moment is $(1-\varepsilon)m_1(\nu_N) + \varepsilon \sum_i F_i p_i = m_1(\nu_N)$ when $m_1(\nu_N) = m_1(\nu_N^\star)$, so $\nu_N^\varepsilon \in \Pi_{\mathcal{M}}(y)$ in both cases. Each mass moves by at most $\varepsilon \operatorname{diam}(P_i)$, so by Lemma 3.13(ii), $|h_{\mathcal{X}(\nu_N^\varepsilon)}(q) - h_{\mathcal{X}(\nu_N)}(q)| \le \varepsilon\mu|b| \sum_i F_i \operatorname{diam}(P_i)$. Let $\varepsilon \to 0$. □

For general convex taxels, the same argument applies whenever every point of $\overline{P_i}$ can be moved into P_i toward a point of P_i; otherwise the bounds below remain valid, possibly conservative, by Theorem 3.16 applied to the larger closure family.

Lemma 5.7 (Concentration and spreading)**.** *Under Assumption 5.1, let $\nu_N = \sum_i \nu_i \in \overline{\Pi}_{\mathcal{M}_F}(y_F)$, with local first moments $u_i \triangleq \int x \, d\nu_i(x)$.*

(i) Concentration. For $F_i > 0$, $c_i \triangleq u_i/F_i \in \overline{P_i}$, and the measure $\nu_N^c \triangleq \sum_{i:F_i>0} F_i \delta_{c_i}$ belongs to $\overline{\Pi}_{\mathcal{M}_F}(y_F)$, has the first moment of ν_N, and

$$\mathcal{X}(\nu_N^c) \subseteq \mathcal{X}(\nu_N). \tag{41}$$

(ii) Spreading. If the closures are polygons, there are corner weights $\lambda_{ik} \ge 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_{i,k} \lambda_{ik} v_{ik} = m_1(\nu_N)$ such that the corner-supported measure $\nu_N^s \triangleq \sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ satisfies

$$\mathcal{X}(\nu_N) \subseteq \mathcal{X}(\nu_N^s). \tag{42}$$

In words: concentrating the force of each taxel at its local CoP shrinks the wrench set, and spreading it to the taxel corners enlarges it, both without changing the taxel forces or the first moment.

Proof. We use the description of $\mathcal{X}(\nu_N)$ by tangential measures ν_T with $\|\nu_T(E)\|_2 \le \mu\nu_N(E)$ (footnote to (13)). (i) $c_i \in \overline{P_i}$ since $\overline{P_i}$ is closed and convex. A wrench of ν_N^c is given by vectors $\gamma_i \in B(0,\mu)$ at the atoms: $(\sum_i F_i\gamma_i, \sum_i c_i \times F_i\gamma_i)$. The tangential measure $\nu_T = \sum_i \gamma_i \nu_i$ is admissible for ν_N, with resultant force $\sum_i F_i\gamma_i$ and moment $\sum_i u_i \times \gamma_i$, the same wrench. (ii) Choose measurable barycentric coordinates $\alpha_{ik}(x) \ge 0$ on $\overline{P_i}$, with $\sum_k \alpha_{ik} = 1$ and $\sum_k \alpha_{ik}(x) v_{ik} = x$ (e.g. bilinear coordinates on a rectangle), and let $\lambda_{ik} = \int \alpha_{ik} \, d\nu_i$, which has the stated sums. A wrench of ν_N is given by an admissible field γ_T; set $\rho_{ik} = \int \alpha_{ik}\gamma_T \, d\nu_i$. Then $\|\rho_{ik}\|_2 \le \mu\lambda_{ik}$, $\sum_{i,k} \rho_{ik} = f_T$ and $\sum_{i,k} v_{ik} \times \rho_{ik} = \sum_i \int (\sum_k \alpha_{ik}(x) v_{ik}) \times \gamma_T(x) \, d\nu_i(x) = \tau_N$, so the corner forces ρ_{ik} realize the same wrench for ν_N^s. □

Spreading is the reduction to the extreme points of the patch in Yann's draft (his Eq. (33) and Proposition 4.4), applied taxel by taxel and at the level of sets.

Remark 5.8 (Intuition)**.** *The infinite-dimensional problems over measures reduce to finite-dimensional ones through Lemma 5.7 (Figure 2).*

- *Lower bounds = "concentrate". One point load per taxel, at its local CoP, is the least favorable distribution with given taxel forces and first moment: every wrench it can carry is carried by the original measure. Needed for all lower bounds: $\mathcal{M}_0$ (Theorem 4.2, a single point at the CoP), $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.1).*
- *Upper bounds = "spread to corners". Splitting each mass among the corners of its taxel, with weights whose average position is the mass's position, keeps the taxel forces and the first moment, and every wrench of the original measure remains available. Needed for all upper bounds: $\mathcal{M}_0$ (Corollary 4.5, corners of P), $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.2).*

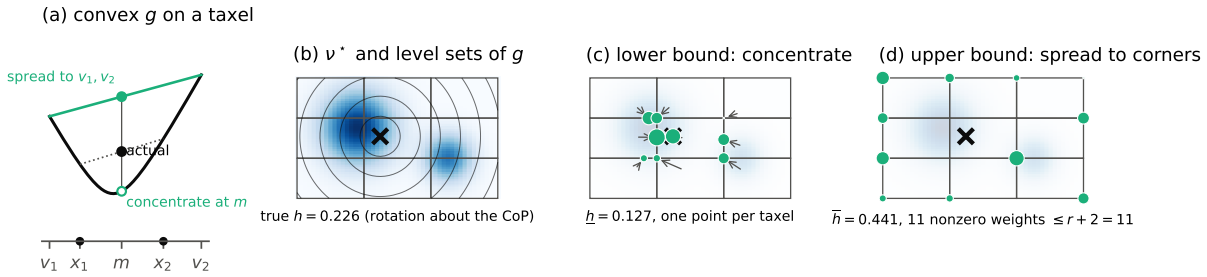


Figure 2: Concentrate vs. spread (Remark 5.8), 3×3 taxels, unit rotation about the CoP $q_c = (-Jc, 1)$, so $g(x; q_c) = \|x - c\|_2$. (a) For masses at x_1, x_2 with CoP m: concentrating at m lowers the objective (Jensen), spreading to the vertices v_1, v_2 raises it (chord). (b) True measure and level sets of g. (c) Minimizer of Theorem 7.1: one Dirac per taxel, pulled from the local CoPs of the taxels toward the CoP (arrows). (d) Maximizer of Theorem 7.2: weights on the taxel corners, pushed away from the CoP, with at most $r + 2$ nonzero weights. Marker areas are proportional to force.

- *Boundaries. A corner of a half-open taxel may belong to its neighbor; Lemma 5.6 shows that corner-supported measures nevertheless give the exact values.*

Remark 5.9 (Compatible CoPs)**.** *In the closure model, the first moments compatible with y_F form the Minkowski sum $\sum_i F_i \overline{P_i}$, by Lemma 5.7(i) and its converse ($\sum_i F_i \delta_{c_i}$ is compatible for any $c_i \in \overline{P_i}$). For $f_N > 0$ the compatible CoPs form the weighted Minkowski average $\sum_i \frac{F_i}{f_N} \overline{P_i}$, a convex set of diameter at most $\sum_i \frac{F_i}{f_N} \operatorname{diam}(P_i)$. The bounds do not use this set; it is shown in the figures, and an observation y_T is consistent only if $m_1 \in \sum_i F_i \overline{P_i}$.*

6 $\mathcal{M}_F$ case (taxel forces)

Known: the taxel forces F_i. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel, hence also its CoP. Without the first-moment constraint, the taxels decouple and both bounds have closed forms.

Theorem 6.1 (Exact bounds for taxel force measurements)**.** *Under Assumption 5.1 with polygonal closures and half-open dyadic cells (or the closure model), for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \min_{x \in \overline{P_i}} \|a + bJx\|_2, \qquad \overline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \max_{k} \|a + bJv_{ik}\|_2.} \tag{43}$$

For $b \neq 0$, with the instantaneous center of rotation $p = Ja/b$, these are $\mu|b| \sum_i F_i \operatorname{dist}(p, \overline{P_i})$ and $\mu|b| \sum_i F_i \max_k \|v_{ik} - p\|_2$. For a rectangular taxel with center m_i and sides $w_x \times w_y$, and $\Delta = m_i - p$,

$$\min_{x \in \overline{P_i}} \|x - p\|_2 = \left\| \left((|\Delta_1| - \tfrac{w_x}{2})_+ , (|\Delta_2| - \tfrac{w_y}{2})_+ \right) \right\|_2 , \tag{44}$$

$$\max_{x \in \overline{P_i}} \|x - p\|_2 = \left\| \left(|\Delta_1| + \tfrac{w_x}{2}, |\Delta_2| + \tfrac{w_y}{2} \right) \right\|_2 . \tag{45}$$

For $b = 0$ both bounds equal $\mu f_N \|a\|_2$.

Proof. By Lemma 5.6, we may optimize over $\overline{\Pi}_{\mathcal{M}_F}(y_F)$, which is the product of the families $\{\nu_i \text{ on } \overline{P_i},\ \nu_i(\overline{P_i}) = F_i\}$, and the objective $\mu \sum_i \int g \, d\nu_i$ is a sum of one term per family. For

each term, $F_i \min_{\overline{P_i}} g \le \int g\, d\nu_i \le F_i \max_{\overline{P_i}} g$, with equality for a point load at a minimizer or a maximizer; the maximum of the convex function $g(\cdot; q)$ over a polygon is attained at a vertex. The rectangle formulas are the distance from p to the rectangle and to its farthest corner. □

Remark 6.2 (Relation with the earlier formulation)**.** *The bounds (43) are the "cellwise" bounds of the combined measurement (the $\mathcal{M}_T$ problem without its first-moment constraint), which are exact for $\mathcal{M}_F$. An earlier version of this note took the torque about the CoP of each candidate measure. Under $\mathcal{M}_F$ the CoP is unknown, and that formulation leads to a bilevel, generally nonconvex problem (an outer optimization over the compatible CoPs of Remark 5.9). Its values are valid bounds on the support of the true set about the true CoP, but they compare twists written about different points for different candidates (Remark 3.5), and cannot be used to test a load given in a fixed frame. They are not used.*

7 $\mathcal{M}_T$ case (taxel forces and CoP)

Known: the taxel forces F_i and the first moment m_1, i.e., the CoP. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel. The taxel constraints act on each taxel separately, but the first-moment constraint $\int_P x\, d\nu_N = m_1$ couples the taxels.

Theorem 7.1 (Exact lower bound for combined measurements)**.** *Under Assumption 5.1 with half-open dyadic cells (or the closure model), given $y_T = (f_N, m_1, F_1, \dots, F_r) = \mathcal{M}_T(\nu_N^\star)$, for every $q = (a, b)$,*

$$\boxed{\begin{aligned} \underline{h}(q; y_T, \mathcal{M}_T) = \mu \min_{u_1,\dots,u_r \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} \|F_i a + bJu_i\|_2 \\ \text{subject to} \quad & u_i \in F_i \overline{P_i}, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} u_i = m_1. \end{aligned}} \tag{46}$$

The variables are the local first moments; for $F_i > 0$, $u_i = F_i c_i$ with $c_i \in \overline{P_i}$ the local CoP, and the objective is $\mu \sum_i F_i\, g(c_i; q)$, while $F_i = 0$ forces $u_i = 0$. The problem is convex, and an SOCP when the closures are polygons $\{x : A_i x \le d_i\}$, since $u_i \in F_i \overline{P_i}$ reads $A_i u_i \le F_i d_i$.

Proof. By Lemma 5.6, we may use the closure model. ($\ge$) For $\nu_N = \sum_i \nu_i \in \overline{\Pi}_{\mathcal{M}_T}(y_T)$, the local first moments u_i are feasible, and $\int g\, d\nu_i \ge \|F_i a + bJu_i\|_2$ by Lemma 3.13(iii) applied to ν_i. ($\le$) For feasible u_i, the concentrated measure $\sum_{i:F_i>0} F_i \delta_{u_i/F_i}$ is compatible and attains the objective. The feasible set is compact and nonempty (it contains the local first moments of $\nu_N^\star$), and the objective is convex and continuous. □

Theorem 7.2 (Exact upper bound for combined measurements)**.** *Under the assumptions of Theorem 7.1, with polygonal closures $\overline{P_i} = \operatorname{conv}\{v_{i1}, \dots, v_{iK_i}\}$, for every $q = (a, b)$,*

$$\boxed{\begin{aligned} \overline{h}(q; y_T, \mathcal{M}_T) = \mu \max_{\lambda_{ik} \ge 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} \|a + bJv_{ik}\|_2 \\ \text{subject to} \quad & \sum_{k=1}^{K_i} \lambda_{ik} = F_i, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} v_{ik} = m_1. \end{aligned}} \tag{47}$$

This is a linear program, and an optimal solution exists with at most $r + 2$ nonzero weights.

Proof. By Lemma 5.6, we may use the closure model. ($\leq$) For $\nu_N \in \overline{\Pi}_{\mathcal{M}_T}(y_T)$, Lemma 5.7(ii) gives a feasible λ with $h_{\mathcal{X}(\nu_N)}(q) \leq h_{\mathcal{X}(\nu_N^s)}(q)$, which is the objective of λ. ($\geq$) For a feasible λ, the corner-supported measure $\sum_{i,k} \lambda_{ik}\delta_{v_{ik}}$ is compatible and attains the objective. The LP is feasible (by spreading $\nu_N^\star$) and bounded, and a basic optimal solution has at most as many nonzero entries as equality constraints, i.e., $r+2$. □

Remark 7.3 (Recovering the coarse and taxel-force cases)**.** *For a single taxel $P_1 = P$, the constraint of (46) forces $u_1 = m_1$, and Theorem 7.1 reduces to Theorem 4.2; Theorem 7.2 reduces to Corollary 4.5, with $r+2=3$ nonzero weights. Dropping the first-moment constraint in (46) and (47) decouples the taxels and gives the $\mathcal{M}_F$ bounds of Theorem 6.1. Consistently with Theorem 3.16 and $\mathcal{M}_T \succeq \mathcal{M}_F$, the $\mathcal{M}_T$ bounds lie between the $\mathcal{M}_F$ bounds:*

$$\underline{h}(q; y_F, \mathcal{M}_F) \leq \underline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_F, \mathcal{M}_F). \tag{48}$$

The difference is the value of measuring the CoP in addition to the taxel forces.

8 Comparison of measurements and tightening under refinement

All results of this section follow from Theorem 3.16, the information chains of Proposition 5.5, and the closed forms of Theorem 6.1.

Proposition 8.1 (Comparison of coarse, taxel and combined measurements)**.** *For every twist $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\begin{aligned}\max\left\{\underline{h}(q; y_0, \mathcal{M}_0),\, \underline{h}(q; y_F, \mathcal{M}_F)\right\} \leq \underline{h}(q; y_T, \mathcal{M}_T) \leq h_{\mathcal{X}(\nu_N^\star)}(q),\\ h_{\mathcal{X}(\nu_N^\star)}(q) \leq \overline{h}(q; y_T, \mathcal{M}_T) \leq \min\left\{\overline{h}(q; y_0, \mathcal{M}_0),\, \overline{h}(q; y_F, \mathcal{M}_F)\right\},\end{aligned}} \tag{49}$$

with $\underline{h}(q; y_0, \mathcal{M}_0) = \mu\|f_N a + bJm_1\|_2$. The bounds of $\mathcal{M}_0$ and $\mathcal{M}_F$ are not ordered in general, in either direction.

Proof. $\mathcal{M}_T = \mathcal{M}_0 \vee \mathcal{M}_F$, Proposition 3.15(iii) and Theorem 3.16. For the last claim: with a single taxel, $\mathcal{M}_F$ measures only f_N, and its bounds are wider than those of $\mathcal{M}_0$ (e.g. $\underline{h}(q; y_F, \mathcal{M}_F) = \mu f_N |b| \operatorname{dist}(p, P) < \underline{h}(q; y_0, \mathcal{M}_0) = \mu f_N |b| \|c - p\|_2$ when the CoP c is not the point of P closest to p). With fine taxels, the $\mathcal{M}_F$ bounds converge to the truth (Proposition 8.3), while the $\mathcal{M}_0$ bounds do not in general (Example 4.7). □

Proposition 8.2 (Tightening under taxel refinement)**.** *Let $\mathcal{M}'_F$ and $\mathcal{M}'_T$ be the taxel-force and combined measurements of a partition $\{P'_j\}$ that refines $\{P_i\}$, e.g. $G(p'_x, p'_y)$ and $G(p_x, p_y)$ with $p'_x \geq p_x$, $p'_y \geq p_y$. Then, for every q and $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, with $y = \mathcal{M}(\nu_N^\star)$ and $y' = \mathcal{M}'(\nu_N^\star)$,*

$$\boxed{\underline{h}(q; y, \mathcal{M}) \leq \underline{h}(q; y', \mathcal{M}') \leq h_{\mathcal{X}(\nu_N^\star)}(q) \leq \overline{h}(q; y', \mathcal{M}') \leq \overline{h}(q; y, \mathcal{M}),} \tag{50}$$

and the guaranteed sets grow and the possible sets shrink.

Proof. Proposition 5.5(i) and Theorem 3.16. □

The monotonicity is associated with nested partitions, not with the number of taxels r: a 3×3 and a 4×4 grid carry different information (Proposition 5.5(iii)), and nothing guarantees that the second gives tighter bounds. The gap bounds below, on the other hand, only require small taxel diameters.

Proposition 8.3 (Resolution-dependent gap bounds)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$ and $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$,*

$$\begin{aligned} 0 \le \overline{h}(q; y, \mathcal{M}) - \underline{h}(q; y, \mathcal{M}) &\le \mu \sum_{i=1}^{r} F_i \left(\max_{\overline{P_i}} g(\cdot; q) - \min_{\overline{P_i}} g(\cdot; q) \right) \\ &\le \mu |b| \sum_{i=1}^{r} F_i \operatorname{diam}(P_i) \le \mu |b| f_N \max_i \operatorname{diam}(P_i), \end{aligned} \tag{51}$$

with equality in the first inequality for $\mathcal{M}_F$. Hence, along any sequence of taxel partitions with $\max_i \operatorname{diam}(P_i) \to 0$, the four bounds of $\mathcal{M}_F$ and $\mathcal{M}_T$ converge to $h_{\mathcal{X}(\nu_N^\star)}(q)$.

Proof. For $\mathcal{M}_F$ the gap is the difference of the two closed forms of Theorem 6.1; for $\mathcal{M}_T$ it is smaller, by (48). By Lemma 3.13(ii), $\max_{\overline{P_i}} g - \min_{\overline{P_i}} g \le |b| \operatorname{diam}(P_i)$. Convergence follows since $h_{\mathcal{X}(\nu_N^\star)}(q)$ lies between the bounds. □

With the torque taken about the CoP of each candidate measure, the corresponding bound for $\mathcal{M}_F$ had an additional factor 2, which came from the unknown reference point.

The next result answers how much one refinement step tightens the bounds. For a twist with $b \neq 0$, let $p = Ja/b$ be its instantaneous center of rotation and $u(x) = (x - p)/\|x - p\|_2$ the unit radial direction at $x \neq p$.

Theorem 8.4 (First-order rate of tightening)**.** *Let P be a rectangle partitioned into a uniform grid of closed-open cells of size $w_x \times w_y$ (e.g. $G(p_x, p_y)$), with cell diameter $D = \sqrt{w_x^2 + w_y^2}$, and let $\nu_N^\star$ have a density $\rho_N \le \bar{\rho}$ with respect to area. Then, for every twist $q = (a, b)$ with $b \neq 0$,*

$$\left| \overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) - \mu |b| \int_P \left(|u_1(x)|\, w_x + |u_2(x)|\, w_y \right) d\nu_N^\star(x) \right| \le \tag{52}$$

$$\mu |b|\, \bar{\rho}\, D^2 \left(\tfrac{15}{4} \sqrt{\pi |P|} + \tfrac{9\pi}{4} D \right), \tag{53}$$

where $|P|$ is the area of P. In particular, for square cells of side h,

$$\overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) = \mu |b|\, C_\nu(p)\, h + O(h^2), \tag{54}$$

$$C_\nu(p) \triangleq \int_P \left(|u_1| + |u_2| \right) d\nu_N^\star \in \left[f_N, \sqrt{2} f_N \right], \tag{55}$$

so that, if $f_N > 0$, each joint dyadic refinement $G(p, p) \to G(p + 1, p + 1)$ divides the $\mathcal{M}_F$ gap by $2 + O(h)$.

Proof. By Theorem 6.1, the gap is $\mu |b| \sum_C F_C e_C$, where the sum is over the cells, $F_C = \nu_N^\star(C)$, and $e_C = \max_{\overline{C}} d - \min_{\overline{C}} d$ with $d(x) = \|x - p\|_2$. Let m be the center of a cell C, $z = m - p$, and $\ell_C \triangleq |u_1(m)| w_x + |u_2(m)| w_y$, which is the range of the linear function $x \mapsto \langle u(m), x \rangle$ over $\overline{C}$.

Far cells ($\|z\|_2 \ge D$). For $x \in \overline{C}$, write $y = x - p$ and $d(x) = \langle u(m), y \rangle + r(x)$. Since the component of y orthogonal to $u(m)$ is that of $x - m$, $0 \le r(x) = \frac{\|y\|_2^2 - \langle u(m), y \rangle^2}{\|y\|_2 + \langle u(m), y \rangle} \le \frac{(D/2)^2}{\|z\|_2}$, because $\|x - m\|_2 \le D/2$ and $\|y\|_2, \langle u(m), y \rangle \ge \|z\|_2 - D/2 \ge \|z\|_2/2$. The maximum and the minimum of d over $\overline{C}$ are therefore within $D^2/(4\|z\|_2)$ of those of the linear part, so $|e_C - \ell_C| \le D^2/(4\|z\|_2)$. Moreover, for $x \in C$, $\|u(x) - u(m)\|_2 \le 2\|x - m\|_2/\|z\|_2 \le D/\|z\|_2$, so $|\ell_C - \ell(x)| \le D^2/\|z\|_2$, with $\ell(x) \triangleq |u_1(x)| w_x + |u_2(x)| w_y$, and $\|z\|_2 \ge \frac{2}{3}\|x - p\|_2$. Hence $|F_C e_C - \int_C \ell \, d\nu_N^\star| \le \frac{15}{8} D^2 \int_C \frac{d\nu_N^\star(x)}{\|x - p\|_2}$.

Near cells ($\|z\|_2 < D$). Both e_C and $\ell(x)$ lie in $[0, D]$ (Lemma 3.13(ii) and Cauchy–Schwarz), so $|F_C e_C - \int_C \ell \, d\nu_N^\star| \le D F_C$. These cells lie in the disk $B(p, \frac{3}{2} D)$, which carries at most $\bar{\rho}\, \frac{9\pi}{4} D^2$.

Summing, the left-hand side of (52) is at most $\mu|b|\left(\frac{15}{8}D^2\bar{\rho}\int_P \frac{dA}{\|x-p\|_2} + \frac{9\pi}{4}\bar{\rho}D^3\right)$, and $\int_P \frac{dA}{\|x-p\|_2} \leq 2\sqrt{\pi|P|}$, the value for the disk of area $|P|$ centered at p. For square cells, $\ell(x) = h(|u_1| + |u_2|)$ and $1 \leq |u_1| + |u_2| \leq \sqrt{2}$ for a unit vector. The ratio of two consecutive gaps is $\frac{C_\nu(p)h/2+O(h^2)}{C_\nu(p)h+O(h^2)}$ with $C_\nu(p) \geq f_N > 0$. □

Remark 8.5 (Reading the rate)**.**

- *In terms of the diameter $D = \sqrt{2}h$ of square cells, the $\mathcal{M}_F$ gap is $\frac{C_\nu(p)}{\sqrt{2}f_N}\mu|b|f_N D + O(D^2)$, between $\frac{1}{\sqrt{2}}$ and 1 times the bound of Proposition 8.3: that bound is sharp up to a factor at most $\sqrt{2}$. The constant $C_\nu(p)$ depends on how the load is distributed around the center of rotation, and is largest when the load is seen from p along the diagonals of the grid.*
- *Refining in one direction only, $G(p_x, p_y) \to G(p_x + 1, p_y)$, halves the term $\int |u_1|\, d\nu\, w_x$ only: the gain depends on the direction of the load seen from p.*
- *The bounded density excludes point loads; for measures with concentrated loads near the cell boundaries, one refinement step can leave the gap almost unchanged at coarse resolution.*
- *The same first-order term bounds the $\mathcal{M}_T$ gap from above, by (48). How much the first-moment constraint reduces the gap at high resolution, i.e., the value of measuring the CoP in addition to the taxel forces, is left open here and quantified numerically in the experimental report.*

9 Certified load factor

A controller does not need the bounds direction by direction. It needs to know whether a given load w can be carried, whether it certainly cannot, and by how much the contact should be strengthened (e.g., by squeezing harder). This is a question about the ray $\{sw : s \geq 0\}$: how far along it the guaranteed and the possible sets extend. This section shows that both answers are convex programs for the taxel measurements, without constructing the sets.

Reference point of the load. The load acting on a grasped object is known about a point o fixed to the object or to the hand. By Lemma 3.4, $sw \in \mathcal{X}_o(\nu_N)$ if and only if $sT_o^{-1}w \in \mathcal{X}(\nu_N)$, so a load about o can always be moved to the origin. We keep o explicit below, since the programs are written most simply with it; the support functions about o are $h_{\mathcal{X}_o(\nu_N)}(q) = \mu\int_P \|a + bJ(x - o)\|_2\, d\nu_N$ and we write $\underline{h}_o$, $\overline{h}_o$, $\mathcal{X}_o^{\pm}$ for the corresponding bounds and sets.

Definition 9.1 (Load factors)**.** *For a nonzero load $w = (f, \tau) \in \mathbb{R}^2 \times \mathbb{R}$, expressed about o, the load factor of ν_N is $s_o^\star(\nu_N; w) \triangleq \max\{s \geq 0 :\ sw \in \mathcal{X}_o(\nu_N)\}$. The certified (lower) and the upper load factors are*

$$\underline{s}(w; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w), \tag{56}$$

$$\overline{s}(w; y, \mathcal{M}) \triangleq \max\left\{s \geq 0 :\ sw \in \operatorname{cl}\mathcal{X}_o^+(y, \mathcal{M})\right\} \geq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w). \tag{57}$$

The load w is carried by every compatible measure if $\underline{s}(w; y, \mathcal{M}) \geq 1$, and by none if $\overline{s}(w; y, \mathcal{M}) < 1$. Since $\mathcal{X}_o(\lambda\nu_N) = \lambda\mathcal{X}_o(\nu_N)$, a contact whose normal force is scaled by λ has load factors scaled by λ.

Proposition 9.2 (Certified and upper load factors)**.** *Let $w = (f, \tau) \neq 0$ and $o \in \mathbb{R}^2$.*

(i) For every measurement operator satisfying Assumption 3.9,

$$\boxed{\begin{aligned}&\underline{s}(w;y,\mathcal{M}) = \max\left\{s\geq 0:\ sw\in\mathcal{X}_o^-(y,\mathcal{M})\right\} = \inf_{q:\,\langle q,w\rangle=1} \underline{h}_o(q;y,\mathcal{M}),\\ &\overline{s}(w;y,\mathcal{M}) = \inf_{q:\,\langle q,w\rangle=1} \overline{h}_o(q;y,\mathcal{M}).\end{aligned}} \tag{58}$$

(ii) Under Assumption 5.1 with polygonal closures $\overline{P_i}=\{x: A_ix\leq d_i\}$, for $\sigma\in\{-1,+1\}$ let

$$\boxed{\begin{aligned}V_\sigma \triangleq \mu \inf_{a\in\mathbb{R}^2,\, b\in\mathbb{R},\, u_i\in\mathbb{R}^2} \quad & \sum_{i=1}^{r} F_i \left\|a + J(u_i - b\,o)\right\|_2 \\ \text{subject to} \quad & a^\top f + b\tau = 1, \qquad \sigma b \geq 0, \\ & \sigma\left(A_iu_i - b\,d_i\right)\leq 0, \qquad i=1,\dots,r, \\ & \textstyle\sum_{i=1}^r F_iu_i = b\,m_1 \qquad (\mathcal{M}_T\ \textit{only}),\end{aligned}} \tag{59}$$

with $V_\sigma=+\infty$ if infeasible. Then $\underline{s}(w;y,\mathcal{M})=\min\{V_{+1},V_{-1}\}$ for $\mathcal{M}\in\{\mathcal{M}_F,\mathcal{M}_T\}$, and each V_σ is an SOCP.

(iii) Under the same assumptions, with vertices v_{ik}, the upper load factor is the SOCP

$$\boxed{\begin{aligned}&\overline{s}(w;y,\mathcal{M}_F) = \min_{a,b,t}\ \mu\sum_{i=1}^{r} F_it_i \quad s.t.\ \ a^\top f + b\tau = 1,\ \ t_i \geq \|a + bJ(v_{ik}-o)\|_2 \ \ \forall i,k,\\ &\overline{s}(w;y,\mathcal{M}_T) = \min_{a,b,y,z}\ \sum_{i=1}^{r} F_iy_i + z^\top m_1 \\ &\qquad\qquad s.t.\ \ a^\top f + b\tau = 1,\ \ y_i + z^\top v_{ik} \geq \mu\|a + bJ(v_{ik}-o)\|_2 \ \ \forall i,k,\end{aligned}} \tag{60}$$

and equivalently the primal SOCP

$$\overline{s}(w;y,\mathcal{M}) = \max_{s,\,\lambda_{ik}\geq 0,\,\rho_{ik}\in\mathbb{R}^2}\ s \quad s.t. \quad \textstyle\sum_k \lambda_{ik} = F_i, \quad \|\rho_{ik}\|_2 \leq \mu\lambda_{ik}, \tag{61}$$

$$\textstyle\sum_{i,k}\rho_{ik} = sf, \quad \sum_{i,k}(v_{ik}-o)\times\rho_{ik} = s\tau, \tag{62}$$

$$\textstyle\sum_{i,k}\lambda_{ik}v_{ik} = m_1 \quad (\mathcal{M}_T\ \textit{only}), \tag{63}$$

whose optimal λ is a most favorable compatible measure (in the closure model), with friction forces ρ_{ik} at its corners.

(iv) If $(a^\star,b^\star,u^\star)$ is optimal for the smaller of the two programs in (ii), let $c_i=u_i^\star/b^\star$ if $b^\star\neq 0$, and let (c_i) be any point loads with $c_i\in\overline{P_i}$ (and $\sum_i F_ic_i=m_1$ for $\mathcal{M}_T$) otherwise. Then $\hat{\nu}_N=\sum_i F_i\delta_{c_i}$ is compatible and $s_o^\star(\hat{\nu}_N;w)=\underline{s}(w;y,\mathcal{M})$: the certificate is attained by a compatible measure.

(v) (Decision rule.) If $\underline{s}\geq 1$, every compatible normal force measure carries w. If $\overline{s}<1$, none does: whatever the actual distribution, no admissible friction field balances w, and the contact slips (quasi-statically). If $\underline{s}<1\leq\overline{s}$, the measurement does not decide. If $\mathcal{M}\succeq\mathcal{M}'$, then $\underline{s}(w;y',\mathcal{M}')\leq\underline{s}(w;y,\mathcal{M})\leq s_o^\star(\nu_N^\star;w)\leq\overline{s}(w;y,\mathcal{M})\leq\overline{s}(w;y',\mathcal{M}')$.

(vi) (Several contacts.) Let K contacts act on the same rigid object, with independent measurements y_k of their normal force measures ν_N^k, the same μ, and patches identified with subsets of a common plane (common coordinates x, torques about the common normal). Assume that the wrench components out of this plane are balanced without friction. The load w is

then carried if $w \in \mathcal{X}_o(\nu_N^1) \oplus \cdots \oplus \mathcal{X}_o(\nu_N^K)$ *(Minkowski sum). The certified and upper load factors of the grasp are*

$$\boxed{\underline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \underline{h}_o(q; y_k, \mathcal{M}^k), \qquad \overline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \overline{h}_o(q; y_k, \mathcal{M}^k),} \tag{64}$$

and (ii)–(iii) hold with the objectives of the K *contacts summed, a shared* (a, b)*, and separate variables per contact. Since* $\bigoplus_k \mathcal{X}_o^-(y_k) \subseteq \bigcap_{(\nu_N^k)} \bigoplus_k \mathcal{X}_o(\nu_N^k)$*,* $\underline{s}_{\mathrm{grasp}}$ *is at least the load factor of the sum of the guaranteed sets of the individual contacts.*

Proof sketch. (i) Fix ν_N. Since $\mathcal{X}_o(\nu_N)$ is convex, compact and contains 0, $\{s \geq 0 : sw \in \mathcal{X}_o(\nu_N)\} = [0, s_o^\star(\nu_N; w)]$ with $s_o^\star < \infty$. Hence $sw \in \mathcal{X}_o^-(y)$ if and only if $s \leq s_o^\star(\nu_N; w)$ for all compatible ν_N, which gives the first equality. A closed convex set is the set of points satisfying $\langle q, \cdot\rangle \leq h(q)$ for all q [4, Section 13]. For $\langle q, w\rangle \leq 0$ this holds for every $s \geq 0$, since $h_{\mathcal{X}_o(\nu_N)} \geq 0$. Therefore $s_o^\star(\nu_N; w) = \inf_{\langle q,w\rangle>0} h_{\mathcal{X}_o(\nu_N)}(q)/\langle q, w\rangle = \inf_{\langle q,w\rangle=1} h_{\mathcal{X}_o(\nu_N)}(q)$ by positive homogeneity (the constraint set is nonempty since $w \neq 0$). Exchanging the infima over ν_N and over q gives the second equality. For $\overline{s}$, the same argument applies to the closed convex set $\operatorname{cl} \mathcal{X}_o^+(y)$, which contains 0, is bounded by Assumption 3.9, and has support function $\overline{h}_o$ (Proposition 3.11(i)).

(ii) By (i) and Theorems 6.1 and 7.1, $\underline{s}(w; y, \mathcal{M})$ is the infimum of $\mu \sum_i F_i \|a + bJ(c_i - o)\|_2$ jointly over (a, b) with $a^\top f + b\tau = 1$ and over point loads $c_i \in \overline{P_i}$ (with $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$); for $\mathcal{M}_F$ the minimum over c_i is attained at the closed-form minimizers of Theorem 6.1. The objective is bilinear in (b, c_i). Split the feasible set by the sign of b and substitute $u_i = bc_i$. For $\sigma b > 0$, this is a bijection between the feasible c_i and the u_i with $u_i/b \in \overline{P_i}$, i.e., $\sigma(A_i u_i - b\, d_i) \leq 0$, and $\sum_i F_i u_i = b\, m_1$. Moreover $bJ(c_i - o) = J(u_i - b\, o)$, so the objective is preserved. For $b = 0$, the original objective is $\mu f_N \|a\|_2$ for every feasible (c_i), and feasible points exist. In the new variables, $\sigma A_i u_i \leq 0$ forces $u_i = 0$ because $\overline{P_i}$ is bounded, and the objective is again $\mu f_N \|a\|_2$. The objective is a sum of norms of affine functions and the constraints are linear, so epigraph variables give an SOCP.

(iii) By (i) and Theorem 6.1, for $\mathcal{M}_F$, $\overline{s} = \inf \mu \sum_i F_i \max_k \|a + bJ(v_{ik} - o)\|_2$ over $\langle q, w\rangle = 1$, which is the stated program in epigraph form. For $\mathcal{M}_T$, the LP of Theorem 7.2 (with $g(v_{ik}; q)$ replaced by $\|a + bJ(v_{ik} - o)\|_2$) is feasible and bounded, so by LP duality $\overline{h}_o(q) = \min\{\sum_i F_i y_i + z^\top m_1 :\ y_i + z^\top v_{ik} \geq \mu \|a + bJ(v_{ik} - o)\|_2\}$; minimizing jointly over (q, y, z) gives the stated program, whose constraints are second-order cones. For the primal form: by Lemmas 5.6 and 5.7(ii), $\operatorname{cl} \mathcal{X}_o^+(y)$ is the union of the sets $\mathcal{X}_o$ of the corner-supported compatible measures, i.e., the projection of the (compact, convex) feasible set of (λ, ρ) onto the wrench $(\sum \rho, \sum (v - o) \times \rho)$.

(iv) $\hat{\nu}_N$ is compatible (Lemma 5.7(i) and its converse). With $q^\star = (a^\star, b^\star)$, by (i), $s_o^\star(\hat{\nu}_N; w) \leq h_{\mathcal{X}_o(\hat{\nu}_N)}(q^\star) = \mu \sum_i F_i \|a^\star + b^\star J(c_i - o)\|_2 = \underline{s}(w; y, \mathcal{M}) \leq s_o^\star(\hat{\nu}_N; w)$.

(v) Definitions, and Theorem 3.16 for the sets.

(vi) For fixed measures, $\bigoplus_k \mathcal{X}_o(\nu_N^k)$ is convex, compact, contains 0, and has support function $\sum_k h_{\mathcal{X}_o(\nu_N^k)}$. The argument of (i) applies. The compatible family is a product of the individual families, and the k-th term of the sum depends only on ν_N^k, so for fixed q the infimum (resp. supremum) of the sum is the sum of the infima (resp. suprema). For the inclusion, if $w_k \in \mathcal{X}_o^-(y_k)$ for each k, then $w_k \in \mathcal{X}_o(\nu_N^k)$ for every compatible ν_N^k, so $\sum_k w_k \in \bigoplus_k \mathcal{X}_o(\nu_N^k)$. □

Remark 9.3 (On Proposition 9.2)**.** • *Why not a grid of directions.* $\underline{h}_o(\cdot; y)$ *is an infimum of convex functions of* q*, and is not convex in general. Minimizing it over a finite set of directions gives a value* $\geq \underline{s}(w; y, \mathcal{M})$*, which may certify a load that is not guaranteed. Minimizing jointly over* q *and the point loads, after the change of variables* $u_i = bc_i$*, gives the exact value as a convex program. The upper bound* $\overline{h}_o(\cdot; y)$ *is convex (a support function), so* $\overline{s}$ *is a convex minimization from the start.*

- *Coarse case. With a single taxel and $o = c$ ($\mathcal{M}_0$), the constraint forces $c_1 = c$, and (ii) gives $\underline{s} = \mu f_N / \|f\|_2$ if $\tau = 0$ and $\underline{s} = 0$ if $\tau \neq 0$ (take $a = 0$, $b = 1/\tau$). The coarse measurement never certifies a torsional load about the CoP (Corollary 4.3).*
- *Membership tests for general operators. When $\overline{h}$ can be evaluated with a maximizing wrench (a support point of $\mathcal{X}^+$) but no closed form is available, e.g., through moment relaxations of the GMP, the distance from w to $\operatorname{cl} \mathcal{X}^+$ can be computed by the Frank–Wolfe method, whose linear minimization oracle is exactly this support point. For the taxel measurements, (iii) is exact and cheaper.*
- *Taxel boundaries. The values are exact for the half-open taxels of dyadic grids (Lemma 5.6). For other partitions they are exact for the closure model and valid, possibly conservative, for the half-open one (Theorem 3.16).*
- *Scope. The certificate is quasi-static and applies to the measured state. In a feedback loop the load and the contact change between decisions, which requires a separate margin. In (vi), the assumption on the out-of-plane components matters: with a free object, offset normal forces at opposing contacts create a couple about an in-plane axis that must be balanced by friction, which the planar model does not account for.*

References

[1] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, Revised ed., CRC Press, Boca Raton, FL, 2015.

[2] J.-B. Lasserre, *Moments, Positive Polynomials and Their Applications*, Imperial College Press Optimization Series, vol. 1, Imperial College Press, London, 2010.

[3] G. Winkler, "Extreme points of moment sets," *Mathematics of Operations Research*, vol. 13, no. 4, pp. 581–587, 1988.

[4] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, Princeton, NJ, 1970.